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Studierichting: Informatica
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Los op in $\mathbb{C}$:

$$z^4 + 4 = 0$$

$$\Leftrightarrow z^4 + 4z^2 + 4 - 4z^2 = 0$$

$$\Leftrightarrow (z^2 + 2)^2 - (2z)^2 = 0$$

$$\Leftrightarrow (z^2 - 2z + 2)(z^2 + 2z + 2) = 0$$

$$\Leftrightarrow z^2 - 2z + 2 = 0 \quad \text{of} \quad z^2 + 2z + 2 = 0$$

$$z^2 - 2z + 2 = 0$$

$$\Leftrightarrow z = \frac{2 \pm \sqrt{-4}}{2}$$

$$\Leftrightarrow z = \frac{2 \pm 2i}{2}$$

$$\Leftrightarrow z = 1 \pm i$$

$$z^2 + 2z + 2 = 0$$

$$\Leftrightarrow z = \frac{-2 \pm \sqrt{-4}}{2}$$

$$\Leftrightarrow z = \frac{-2 \pm 2i}{2}$$

$$\Leftrightarrow z = -1 \pm i$$

$$z_1 = 1 + i$$

$$z_2 = 1 - i$$

$$z_3 = -1 + i$$

$$z_4 = -1 - i$$

References